21/03

fprintf and fscanf

```
int printf(const char *format,...);
int scanf(const char *format,...);
```

ex: printf("%d %f",a,b); writing to stdin
scanf("%d %f",&a,&b); reading from stdout

```
int fprintf(FILE *stream,const char * format,...);
int fscanf(FILE *stream ,const char * format,...);
```

ex :

```
fprintf(fp,"%d %f",a,b); // writes to the file
fscanf(fp,%d %f",&a,&b); // reads from file
```

feof()---> returns 1 on end of file
returns 0 on non end of file

```
while(!feof(fp))

while(!feof(fp))
{
fgets(str,100,fp);
printf("%s\n",str);
}
```

typedef

typedef is a user defined datatype, which is use to give another name or alias name to datatypes

typedef is used to increase readability of the program
typedef provides protability
typedef with structures
use to give simple and short names, so thast we avoid struct key word inthe varaible, pointers and array declarations

typedef with Pointers

```
typedef int * INTPTR
```

INTPRT p1,p2 ---> both p1 and p2 are integer pointers

typedef unnamed structure

typedef with arrays

```
int arr[10];

typedef int arr[10];
```

↳ arr is a short simple name to array of 10 int

ex: arr b; // same as int b[10]

arr c[3]; // int c[10][3];